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Alberta stroke programme early CT score (ASPECTS)

Last revised by Giorgio Maria Agazzi (/users/giorgiobs90?lang=us) on 19 Sep 2026

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The **Alberta stroke program early CT score (ASPECTS)** [1] is a 10-point quantitative topographic CT score used for patients with middle cerebral artery (MCA) stroke. It has also been adjusted for the posterior circulation (see below).

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Usage

An ASPECTS score ≤7 predicts a worse functional outcome at 3 months as well as symptomatic hemorrhage. According to the study by Aviv et al., patients with an ASPECTS score <8 treated with thrombolysis did not have a good clinical outcome [3].

Classification

Segmental estimation of the middle cerebral artery (MCA) (/articles/middle-cerebral-artery?lang=us) vascular territory is made, and 1 point is deducted from the initial score of 10 for every region involved:

- caudate (/articles/caudate-nucleus?lang=us)
- putamen (/articles/putamen?lang=us)
- internal capsule (/articles/internal-capsule?lang=us)
- insular cortex (/articles/insular-cortex?lang=us)
- M1: "anterior MCA cortex", corresponding to the frontal operculum
- M2: "MCA cortex lateral to insular ribbon" corresponding to the anterior temporal lobe
- M3: "posterior MCA cortex" corresponding to the posterior temporal lobe
- M4: "anterior MCA territory immediately superior to M1"
- M5: "lateral MCA territory immediately superior to M2"
- M6: "posterior MCA territory immediately superior to M3"

Important points:

- the initial paper [1] specifically referred to the internal capsule only involving the posterior limb, however, subsequent articles indicate any portion of the internal capsule may be included
- M1 to M3 are at the level of the basal ganglia
- M4 to M6 are at the level of the ventricles immediately above the basal ganglia
- M1 to M4 are not to be confused with the anatomic segments of the middle cerebral artery (/articles/middle-cerebral-artery?lang=us)

Posterior circulation

Variations of the ASPECT scoring system have been described for use in the posterior circulation and referred to as pc-ASPECTS [5].

As is the case for the anterior circulation, the pc-ASPECTS is a 10 point scale, where points are lost for each region affected. Unlike ASPECTS, the pons and the midbrain are worth 2 points each (regardless of whether or not the changes are bilateral; any involvement of the pons, for example, deducted 2 points).

- thalami (1 point each)
- occipital lobes (1 point each)
- midbrain (2 points)
- pons (2 points)
- cerebellar hemispheres (1 point each)

Limitations

Several limits of ASPECT scoring system are recognized:

- unequal weighting across brain regions: individual ASPECT areas cover different volumes, each deducted point does not correspond to a consistent amount of damaged brain [6]
  - two patients with same ASPECT score can have different volume of infarcted area and very different clinical presentation
  - different regions contribute unequally to prognosis in patients with stroke (M6 showing the worse prognosis) [8]
- the exact amount of damage to consider a region positive has never been defined [6]
- interobserver variability is reported [7]
  - thin slices (<2.5 mm) can help assessing M3 and insular ribbon regions, with insular ribbon showing the lowest agreement in studies

See also

- CTA-SI ASPECTS (/articles/ct-angiography-source-image-aspects?lang=us)

External links

- interactive ASPECTS calculator - Rad At Hand (https://radathand.com/radiology-calculators/neuroradiology/aspects-calculator/)

If any of these links are broken or for other problems and questions, please contact editors@radiopaedia.org (mailto:editors@radiopaedia.org).

+ **Quiz questions**

× **References**

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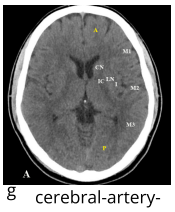
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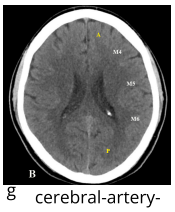
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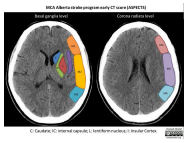
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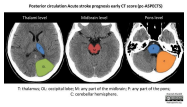
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